

WORKING WITH LARGER DATA IN R

Hands-On Worksheet: A Large Chess Data File

UD Data Analytics Club | September 9, 2026

Goal. Practice making sensible decisions before loading a genuinely large file into R. If we do not finish this activity during the workshop, you can complete it on your own afterward.

Part 1. Know Your Computer

Before downloading a large file, check the computer you are using.

Windows

1. Click Start → Settings.
2. Select System → About.
3. Under Device specifications, find Installed RAM.
4. Under Windows specifications, find your Windows version.

macOS

1. Click the Apple menu () in the upper-left corner.
2. Select About This Mac.
3. Find Memory (RAM) and your macOS version.

Operating system: _____ **Installed RAM:** _____ GB

Now, open RStudio and type the following command in the R console.

```
Sys.info()[c("sysname", "release", "machine")]
R.version$arch
```

Does R's architecture match what you expected from your computer? ☐ Yes ☐ No ☐ Not sure

BEFORE YOU DOWNLOAD: File size is not the same as memory usage. A file that fits on your drive may require more memory after R loads and processes it. When in doubt, choose a smaller file first.

Quick judgment: Suppose your computer has 8 GB of RAM. Would you choose a 7 GB file for this activity? Why or why not?

Part 2. Choose a FICS Chess Data File

Visit the FICS Games Database: <https://www.ficsgames.org/download.html>

Use the download options to create a PGN file that is reasonable for your computer. **Check the estimated file size before downloading.** For this activity, aim for approximately **50–200 MB**. When in doubt, start smaller.

Setting	Your Selection
Game Type	_____
Year	_____
Month	_____
Move Times	☐ Include ☐ Do not include
Estimated Size	_____ MB

Is this a reasonable size for your computer?

☐ Yes ☐ No ☐ Not sure

If your file is very small: Change one setting at a time and observe how the estimated file size changes before downloading.

What setting had the biggest effect on file size?

Part 3. Read and Inspect the File

Place the downloaded PGN file in your R working directory. Replace the filename below with your file.

```
pacman::p_load(data.table)
```

```
chess_file <- "your_file.pgn"
```

Step 1 — Peek before loading everything.

```
chess_sample <- fread(  
  chess_file,  
  header = FALSE,  
  sep = "\n",  
  nrows = 100  
)  
  
head(chess_sample, 20)
```

What do you notice about the structure of the file?

Step 2 — If the file size is reasonable, read the full file.

```
read_time <- system.time({  
  chess_raw <- as.vector(  
    fread(  
      chess_file,  
      header = FALSE,  
      sep = "\n",  
      stringsAsFactors = FALSE  
    )  
  )  
})  
  
read_time  
length(chess_raw)  
format(object.size(chess_raw), units = "MB")
```

Reading time

Number of elements

Size of R object

Part 4. Think Before You Load

Suppose you find a PGN file that is much larger than the one you just used.

1. Would `fread()` make it safe to load a file of any size? Explain.

2. What could happen if the file is too large for the available memory?

3. What could you do instead of loading the entire file into memory at once?

TAKEAWAY: Working with larger data is not just about using a faster function. It is about choosing a workflow that matches the data and your computer.